

KENDRIYA VIDYALAYA KANKARBAGH
FIRST TERM EXAM

Paper ID : QP-1783067622405

Board : CBSE

Class : 12

Subject : Computer Science

Time : 20 minutes

Maximum Marks : 20

Student Name : Vivek Singh

Roll No : 1

Section : A

Date : 03/07/26

GENERAL INSTRUCTIONS

1. All questions are compulsory.
2. Read all questions carefully.
3. Figures in the right margin indicate marks.
4. Draw neat diagrams wherever necessary.
5. Write answers in clear handwriting.
6. Use blue or black ink only.
7. Do not write in the margins.
8. Maintain proper question numbering.

SECTION A

Q1. Which of the following will evaluate to True? Given: python s1 = "hello" s2 = "THERE" s3 = "Goldy" s4 = "HELLO"

[1]

- A. s2.islower()
- B. s3.isupper()
- ☒ C. s1.islower()
- D. s3.islower()

(c) s1.islower()

Q2. What precision do floating-point numbers typically represent in Python?

[1]

- A. Single precision (7 digit precision)
- ☒ B. Double precision (15 digit precision)
- C. Triple precision (20 digit precision)
- D. Variable precision based on value

Q3. Which of the following is a hyperbolic trigonometric function available in NumPy?

[1]

- A. $\cos(x)$
- B. $\arcsin(x)$
- ☒ C. $\sinh(x)$
- D. $\text{round_}(a)$

Q4. Which of the following is an example of a list containing mixed value types?

[1]

- A. [1, 2, 3]
- B. ['one', 'Two', 'Three']
- C. ['a', 'b', 'c']
- ☒ D. [4, 1, 'x', 3.5, 'zero']

Q5. How is a percentage sign printed in a format string without being interpreted as a special character?

[1]

- A. By using a single percentage sign (%)
- B. By escaping it with a backslash (%)
- ☒ C. By using two percentage signs (%%)
- D. By enclosing it in quotes (%)

SECTION B

Q6. Explain the purpose of `pop()` and `remove()` methods for lists. When does `pop()` raise an exception?

[3]

`pop()` removes an element from given position (index) in the list.

Q7. What is the significance of 'matching shape' when plotting two sequences using `pyplot.plot()`?

[2]

Q8. List four types of objects that a Python module can contain.

[2]

docstrings, variable and label for data, constants, classes.

SECTION C

Q9. Write a recursive Python function to compute the greatest common divisor (GCD) of two numbers using the Euclidean algorithm.

[3]

```
def gcd(a, b):  
    if b == 0:  
        return a  
    else:  
        return gcd(b, a % b)
```

- Q10. The following recursive function 'skip_prod(n)' is intended to return the product of 'n * (n-2) * (n-4) * ...'. However, it gives a 'RecursionError' when run with an odd number as a parameter. Figure out the reason for this error and suggest a solution by correcting the code. python def skip_prod(n): if n == 0: return 1 else: return n * skip_prod(n-2) [5]

def skip_prod(n):

*** End of Question Paper ***

Exam Controller

Dr. Indrajit Kumar
Principal